17.2 Asexual Reproduction

Production of an offspring by a single parent without the formation and fusion of gametes is called asexual reproduction. It involves only mitotic cell divisions and meiosis does not occur. Offspring produced by asexual reproduction are not only identical to parents but are also exact copies of their parent.

Asexual reproduction occurs by **spore formation.** This is the most common method of asexual reproduction in fungi and bacteria.

During spore formation a structure called **sporangium** develops from the **fungal hypha**. The nucleus divides several times within the sporangium and each nucleus with small amount of cytoplasm develops into a spore. The spores are liberated and they develop into new hypha after reaching the ground or substratum.



Figure 17.6 Spore formation in *Rhizopus*

Activity 1

- Wet a slice of bread, and keep it in a cool, moist and dark place.
- Observe the surface of the slice with a magnifying glass.
- Record your observations for a week.

17.3 Sexual Reproduction in Plants

Sexual reproduction is the process in which two gametes (male and female) are fused to produce offspring of their own kind. In such cases both sexes, male and female sex organs are needed to produce gametes. You have already learnt that the flower is a reproductive organ of a flowering plant. To understand this further we need to study the structure of a flower.

17.3.1 Parts of a Typical Flower

A flower is a modified shoot with limited growth to carry out sexual reproduction. A flower consists of four whorls borne on a thalamus. These whorls are from outside

a) Calyx – consisting of sepals
b) Corolla – consisting of petals
c) Androecium – consisting of stamens
d) Gynoecium or pistil – consisting of carpels



Figure 17.7 Parts of a flower

Activity 2

- Take a shoe flower from a growing plant.
- Observe the floral parts Calyx, Corolla, Androecium and Gynoecium.
- Separate the stamens and carpels and observe the parts.
- Dust the pollen grains on a slide and observe under a microscope.

The **two outermost whorls calyx** and **corolla** are **non–essential** or **accessory whorls** as they do not directly take part in the reproduction. The other two whorls **androecium** and **gynoecium** are known as the **essential whorls**, because both take part directly in reproduction.

Androecium: Androecium, the **male part** of flower is composed of **stamens**. Each stamen consists of a stalk called the **filament** and a small bag like structure called **anther** at the tip. The pollen grains are produced in the anther within the pollen sac.

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